DEPARTMENT OF CONSTRUCTION ENGINEERING AND SRIJAN ‘26
PRESENTS

DATA DRIFT

Date of Event : 11th April, 2026

Problem Statement

Team member :

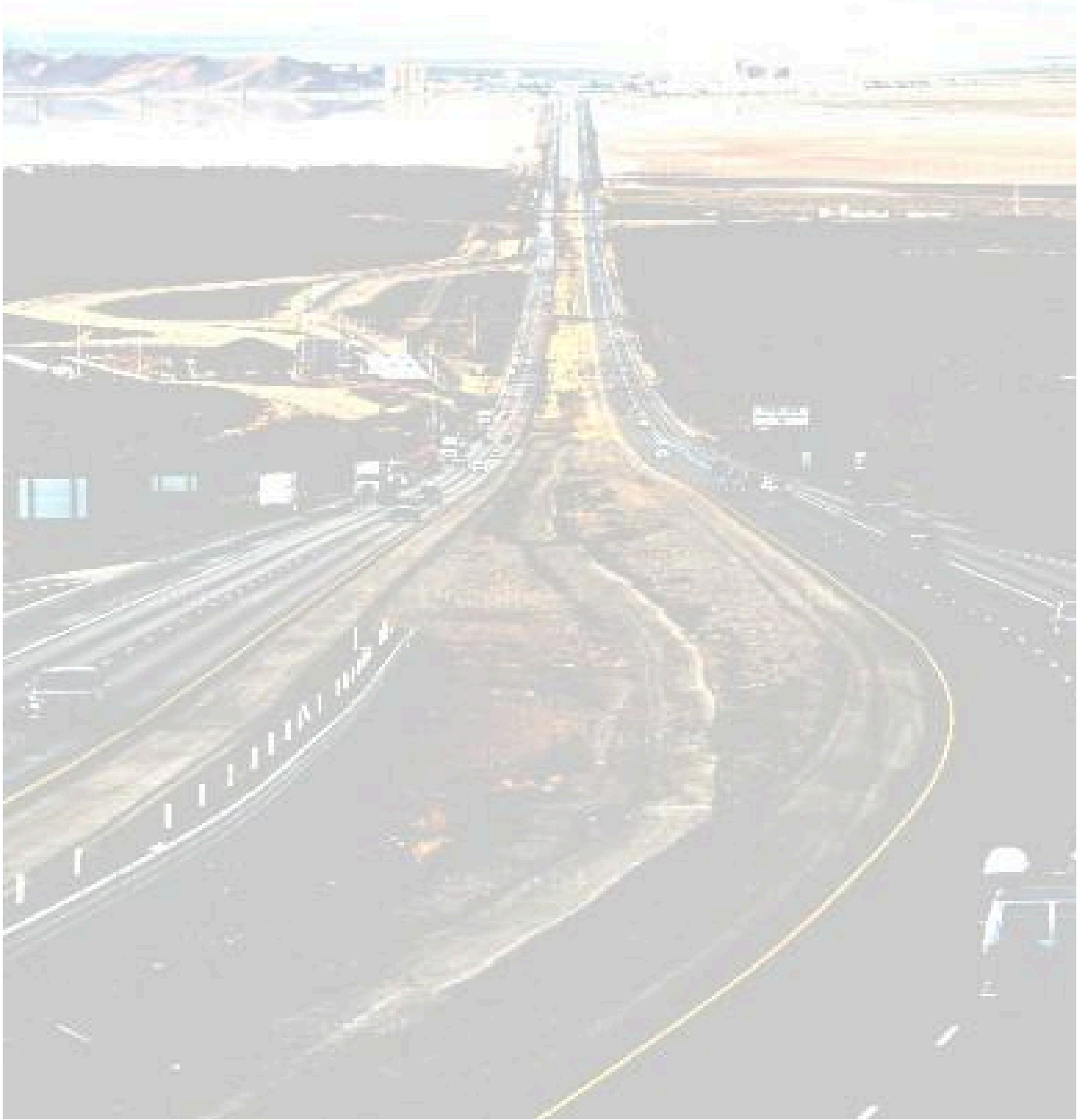
01 (min.) to 02 (max.) Students from various educational institutions and departments of any year of study at the Undergraduate level can work together to form a team.

Introduction :

Smart energy highway systems combine clean energy, sophisticated traffic technologies, and highway engineering to reduce automobile emissions. Analysis of traffic data is used to examine patterns of vehicle movement, delays, and congestion that result in increased pollution and fuel use. Engineers can find inefficient flow circumstances and suggest intelligent control strategies like adaptive signals, sensor-based merge management, and energy-efficient infrastructure by using analytical tools on traffic statistics. The objective is to produce useful information that promotes reduced emissions and more environmentally friendly highway operations.

The dataset provided focuses on highway traffic conditions, capturing data from various road segments over time. It includes key parameters such as vehicle count, average speed, traffic density, weather conditions

(rainfall,visibility), sand emission levels (CO_2, NOx, PM2.5). Imagine you are a data analyst or graduate engineer working in the traffic management department of your city's municipal authority. Your task is to analyze this dataset to understand traffic behavior, identify congestion patterns, evaluate environmental impact, and derive insights that can help improve traffic flow, reduce emissions, and support smarter highway management decisions.



Task :

Data Cleaning : Data cleansing is a crucial step in refining raw datasets into organised and pertinent entries. Within the provided dataset, the following unrefined entities needs to be identified:

1. Duplicate entries: Instances where identical records appear more than once, which can introduce redundancy and inaccuracies in analysis.
2. Blank entries: Records lacking essential information, which can impede comprehensive data analysis and interpretation.
3. Spelling mistakes in some data: Incorrect spellings in the data can cause confusion and mistakes in understanding, which can then lead to errors in later analyses.

Data Processing :

1. Group congestion levels and compare their impact on average speed and emissions across road sections.
2. Add a new feature "hour" by extracting it from the timestamp column and determine the peak traffic hours based on vehicle count.
3. Add a new feature "traffic efficiency index" using speed and density values to evaluate road performance and identify congested segments.

Data Analysis Any 6 out of 8:

1. Identify the peak traffic hour using vehicle count.
2. Calculate average speed per road segment and find the slowest ones.
3. Determine congestion prone zones using traffic density.
4. Analyse CO_2 emissions across congestion levels.
5. Study the impact of rainfall on speed.
6. Identify top 5 emission hotspots using x and y.
7. Analyse heavy vehicles percentage vs congestion level.
8. Identify low visibility zones and their congestion pattern

Advanced Analysis Any 3 out of 5:

1. Analyse impact of gore points on speed and congestion.
2. Simulate increase in EV percentage and its effect on Co2 emissions.
3. Identify non-linear relationship between density and speed.
4. Detect anomalies in traffic patterns.
5. Rank road segments using traffic efficiency index.

KPIs:

1. Total traffic volume.
2. Average Speed.
3. Average Co2 emissions.
4. Emissions per vehicle.
5. Traffic efficiency index.
6. Calculate the KPIs.

Create a Dashboard:

1. Develop an intuitive dashboard interface to consolidate and display key metrics and data points effectively.
2. Utilize visualization tools to present complex information in a visually appealing and accessible format.

Create a Presentation (ppt) with Insights:

1. Construct a structured presentation that synthesizes data-driven insights and analysis.
2. Highlight significant trends, patterns, and correlations discovered through thorough examination of the data.

Finally, you can also extract/analyze the data and find some new insights which are not given in the above tasks. The fundamental intention behind the new findings should be such that it helps the city's traffic planning board to modify the existing policies.

Time :

A total time of 2.5 hours will be provided for analysing the data, creating the dashboard and preparing a one-slider PPT.

Judging criteria :

1. Accurate data cleaning is essential for reliable analysis.
2. Creation of required features such as hour and traffic efficiency index must be correctly implemented.
3. Explore and measure as many KPIs as possible.
4. Innovative thinking and additional insights beyond the given metrics for extra credit.
5. Respond thoughtfully to judges' questions for extra points.
6. Completing the task early may earn bonus points.
7. Use Excel exclusively for creating your dashboard and calculations, ensuring consistency and accuracy.